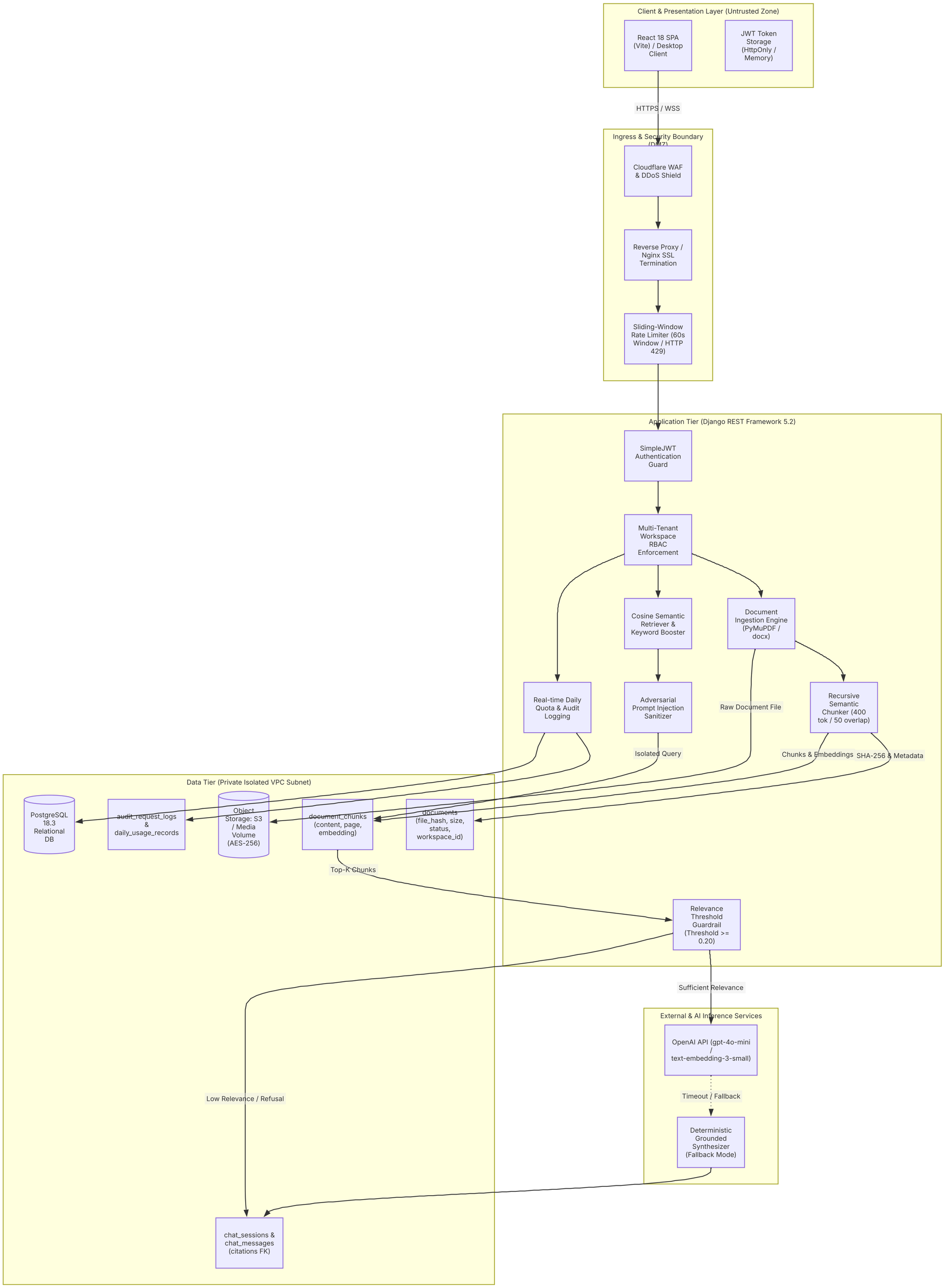

Client & Presentation Layer (Untrusted Zone)
React 18 SPA (Vite) / Desktop Client
JWT Token Storage (HttpOnly / Memory)
HTTPS / WSS
Ingress & Security Boundary (DMZ)
Cloudflare WAF & DDoS Shield
Reverse Proxy / Nginx SSL Termination
Sliding-Window Rate Limiter (60s Window / HTTP 429)
Application Tier (Django REST Framework 5.2)
SimpleJWT Authentication Guard
Multi-Tenant Workspace RBAC Enforcement
Cosine Semantic Retriever & Keyword Booster
Document Ingestion Engine (PyMuPDF / docx)
Real-time Daily Quota & Audit Logging
Adversarial Prompt Injection Sanitizer
Raw Document File
Recursive Semantic Chunker (400 tok / 50 overlap)
Isolated Query
Chunks & Embeddings
SHA-256 & Metadata
Data Tier (Private Isolated VPC Subnet)
PostgreSQL 18.3 Relational DB
audit_request_logs & daily_usage_records
Object Storage: S3 / Media Volume (AES-256)
document_chunks (content, page, embedding)
documents (file_hash, size, status, workspace_id)
Top-K Chunks
Relevance Threshold Guardrail (Threshold >= 0.20)
Sufficient Relevance
External & AI Inference Services
OpenAI API (gpt-4o-mini / text-embedding-3-small)
Timeout / Fallback
Deterministic Grounded Synthesizer (Fallback Mode)
Low Relevance / Refusal
chat_sessions & chat_messages (citations FK)